Chapter One: Introduction

1.1 Background of the organization

Jigjiga University is one of the public higher education institutions established in March 2007 as part of Ethiopia's Second-Generation Universities. It is located in the emerging town of Jigjiga, the capital city of the Ethiopian Somali Regional State, approximately 635 kilometers from Addis Ababa. The University officially commenced its academic services in 2007 with an initial enrollment of 712 students, supported by 66 academic staff and 99 administrative personnel across three faculties.

In recent years, the university's enrollment capacity has grown significantly. Currently, Jigjiga University hosts a total of **21,554 undergraduate students** across **57 academic programs**, along with **627 postgraduate students** studying in **six different postgraduate programs**. These programs operate under **eight Colleges, three Schools, and one Institute**, reflecting the institution's expanding academic structure.

The University has recorded substantial achievements in teaching, research, and community service. In the past year alone, **1,888 students** graduated from various disciplines, contributing to the skilled workforce needed for national development.

Guided by its mission, the University emphasizes demand-driven research and the dissemination of research outputs that are problem-solving and results-oriented. Much of its research focuses on pastoral and agro-pastoral studies, which are highly relevant to the region. One of its strategic priorities is effective research communication, including the adaptation and transfer of locally relevant technologies and raising community awareness on cross-cutting issues. To achieve this, the University engages internal and external stakeholders through diverse platforms such as media outreach, seminars, symposiums, and intellectual forums.

1.2 Statement of the Problem

1.2.1 Existing System

Currently, Jigjiga University does not have a dedicated digital navigation or routing system to guide new students, visitors, or staff around the campus. Freshman students often struggle to locate classrooms, departments, administrative offices, and service centers, especially during their first weeks.
While some departments may provide printed maps or informal guidance, and users may refer to Google Maps, these options are incomplete, outdated, or lack detailed building-level navigation. As a large and expanding institution, the absence of an organized and centralized campus navigation system creates confusion and reduces efficiency.

1.2.2 Major Problems of the Existing System

The shortcomings of the current situation include:

1. **Lack of a centralized and detailed campus map**
   No single platform provides accurate information on buildings, departments, classrooms, and facilities.
2. **Difficulty for new students and visitors**
   Freshman students and guests often get lost, arrive late to classes or appointments, or rely on guesswork and random directions.
3. **Limited and inaccurate online information**
   Google Maps and similar tools show only basic location data, without building names, routes inside campus, or block details.
4. **No real-time or interactive navigation**
   Users cannot get walking routes, step-by-step directions, or updated paths within campus.
5. **Inefficient communication**
   Staff spend unnecessary time giving directions, especially during academic registration or campus events.

These problems reduce productivity, inconvenience users, and negatively impact the overall campus experience.

1.2.3 Proposed System

The proposed system, **UniMap**, is a web-based and optionally mobile-accessible smart campus navigation platform designed to provide interactive, accurate, and user-friendly digital mapping for Jigjiga University.
The system will include:

- An interactive digital map showing buildings, blocks, departments, offices, and key facilities
- Search functionality to quickly locate any building or department
- Step-by-step walking route guidance within the campus
- Building details, images, descriptions, and service information
- Admin panel for updating map data, buildings, and routes
- Optional modules: event locations, emergency points, announcements, and facility management

UniMap aims to make navigation easier, faster, and more reliable for all campus users.

1.2.4 Advantages of the Proposed System

The proposed system offers multiple benefits, including:

1. **Improved navigation and accessibility**
   Students, staff, and visitors can easily find their way around the campus.
2. **Reduced confusion and time wastage**
   Users receive accurate directions, minimizing delays and missed classes or appointments.
3. **Centralized and reliable campus information**
   All building names, departments, and routes are stored in one platform.
4. **Increased efficiency for administrative staff**
   Staff no longer need to repeatedly guide or assist lost individuals.
5. **Enhanced university image and modernization**
   Adopting digital campus navigation aligns with the university's vision for technology-driven solutions.
6. **Scalability**
   The system can be expanded with new buildings, routes, and additional smart campus features.

1.3 Motivation

As Jigjiga University continues to grow in student population, academic programs, and infrastructure, the need for efficient navigation and information access becomes increasingly important. Each year, thousands of freshman students, visitors, and even staff face challenges locating buildings, departments, classrooms, and administrative offices. This often leads to delays, frustration, and reduced productivity.

The motivation for this project arises from the desire to enhance the campus experience through technology-driven solutions. A smart, interactive digital map not only supports smooth mobility but also reflects the university's commitment to modernization, digital transformation, and improved service delivery. Developing UniMap contributes to solving a real-world problem within the university, while also serving as an excellent academic capstone project that demonstrates practical software engineering skills, full-stack development, system design, and user-centered innovation.

1.4 Scope and Limitation of the Project

1.4.1 Scope of the Project

The scope of the proposed UniMap system includes:

1. **Interactive Web-Based Map**
   A digital map showing campus buildings, blocks, roads, pathways, and key service points.
2. **Search and Filtering Features**
   Users can search for departments, offices, buildings, or service areas.
3. **Route Navigation**
   Walking path suggestions from one location to another within the campus.
4. **Building Information Pages**
   Display images, descriptions, and purpose of each building.
5. **Admin Management Panel**
   Authorized personnel can update building names, routes, and map data.
6. **Responsive Design**
   The system will be accessible through web browsers, and optionally packaged as a mobile app.
7. **User Types**
   - General users: students, visitors
   - Admin users: campus map managers or ICT unit

This scope ensures delivery of a functional, user-friendly, and scalable navigation platform.

1.4.2 Limitations of the Project

Despite its usefulness, the system will have the following limitations:

1. **No real-time GPS tracking**
   The system may not support continuous live tracking of user movement due to privacy and resource constraints.
2. **Indoor navigation limitations**
   The project will not include detailed indoor mapping or floor-by-floor navigation inside buildings.
3. **Dependent on available map data**
   The accuracy of the map relies on the information provided by the university; incomplete or outdated data may affect results.
4. **No AI/ML-based smart prediction**
   The system will not use machine learning for automatic path optimization or traffic analysis.
5. **Internet dependency**
   Users need an internet connection to access the web application.

6. **Limited to Jigjiga University campus**
   The system is designed specifically for JU and does not cover external routes or the city of Jigjiga.

1.5 Objective

1.5.1 General Objective

The general objective of this project is to design and develop a smart, interactive, and user-friendly campus navigation system (UniMap) for Jigjiga University that enables students, staff, and visitors to easily locate buildings, departments, and facilities through an accurate digital map and guided routing system.

1.5.2 Specific Objectives

The specific objectives of the project include:

1. To collect and organize accurate geographic and structural information of Jigjiga University campus.
2. To design an interactive digital map that visually represents blocks, buildings, roads, and key locations within the university.
3. To implement a search functionality that allows users to locate buildings, departments, and offices efficiently.
4. To develop a routing feature that provides walking directions from one point on campus to another.
5. To design a responsive web interface accessible on both desktop and mobile devices.
6. To build an administrative panel that enables authorized staff to update map data, building details, and route information.
7. To test and validate the system with real users to ensure usability, accuracy, and performance.
8. To deploy the system on a web server and optionally package it as a mobile application.

1.6 Methodology

1.6.1 Data Collection Methodology

To gather the required information for building the campus map and user requirements, the following data collection methods will be used:

1. **Field Observation**
   Conduct on-site visits to collect the physical layout, building names, GPS coordinates, and pathways.
2. **Interviews & Consultations**
   Engage with administrative staff, security personnel, campus guides, and selected students to understand common navigation challenges.
3. **Document Review**
   Review existing campus maps, architectural layouts, brochures, and any available documents from the registrar or facilities management office.
4. **Online Resources**
   Use satellite imagery and existing tools such as Google Maps or OpenStreetMap to support accuracy where applicable.

1.6.2 System Design and Analysis Tools

The following tools and techniques will be used during system design and analysis:

- **UML Diagrams**
  Including Use Case Diagrams, Activity Diagrams, Sequence Diagrams, and Class Diagrams.
- **ERD (Entity Relationship Diagram)**
  For database structure modeling.
- **Wireframes / UI Mockups**
  Using tools like Figma or Adobe XD to design user interface layouts.
- **Requirement Analysis Techniques**
  Such as functional and non-functional requirement specification.

1.6.3 System Development Tools

The development of the UniMap system will rely on the following technologies and tools:

- **Frontend:**
  React.js / Next.js for building the interactive user interface.
  Tailwind CSS or Material UI for styling.
- **Mapping & Navigation:**
  Leaflet.js or Mapbox for map rendering and route visualization.
  OpenStreetMap as base map data.
- **Backend:**
  Node.js (Express) or Django REST Framework for API development.
- **Database:**
  PostgreSQL or MongoDB for storing map data, locations, and user information.

- **Mobile App Packaging (optional):**
  Capacitor or Expo to wrap the web app and deploy it on Play Store/App Store.
- **Development Tools:**
  VS Code, Git & GitHub for version control, Postman for testing APIs.
- **Deployment:**
  Render, Vercel, Netlify, or AWS for hosting the system.